



Course title: Electronic Engineering

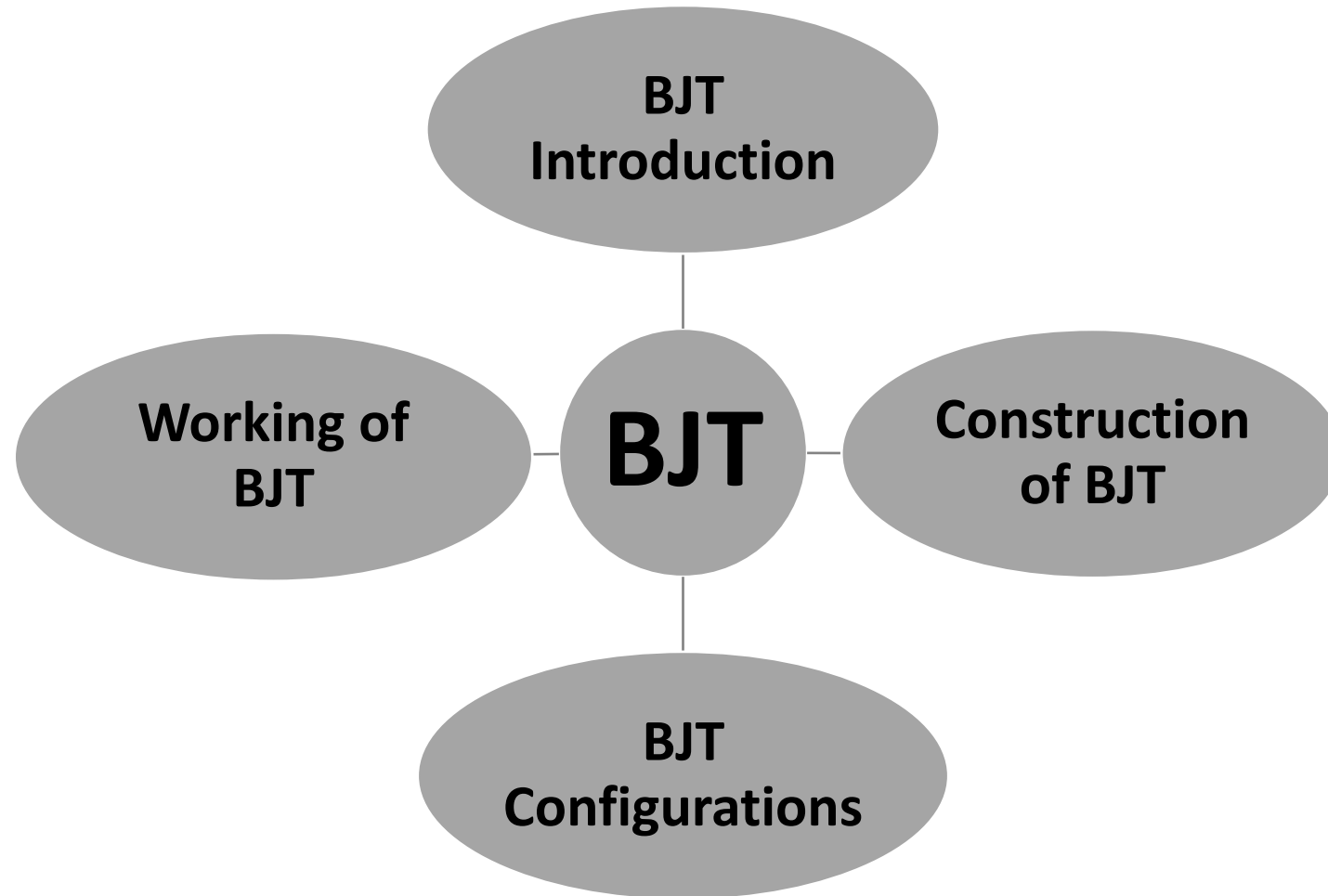
Course Code: CSE 2109



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What will we cover today?



Introduction:

- BJT - Bipolar Junction Transistor
- 3 terminal device
- Both electrons and holes participate in current conduction.
- It is mainly used for signal amplification and electronic switching.



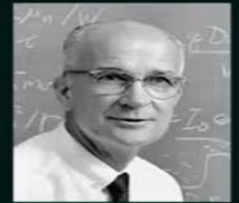
Co-inventors of the first transistor at Bell Laboratories: Dr. William Shockley (seated); Dr. John Bardeen (left); Dr. Walter H. Brattain. (Courtesy of AT&T Archives.)



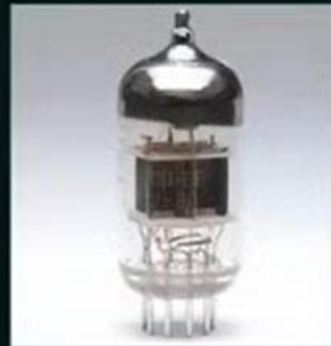
John Bardeen



Walter Brattain



William Shockley



Vaccum Tube



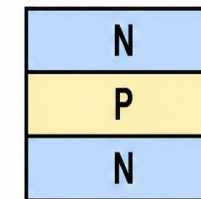
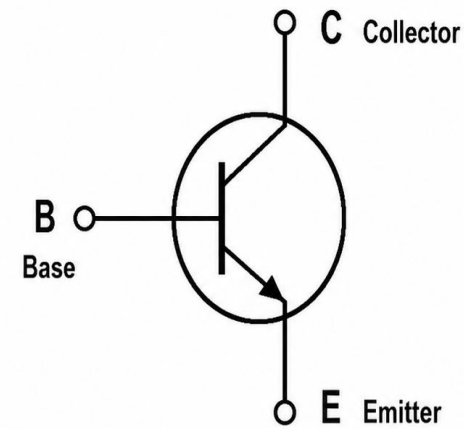
Transistor



Construction of BJT :

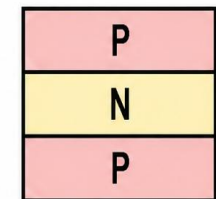
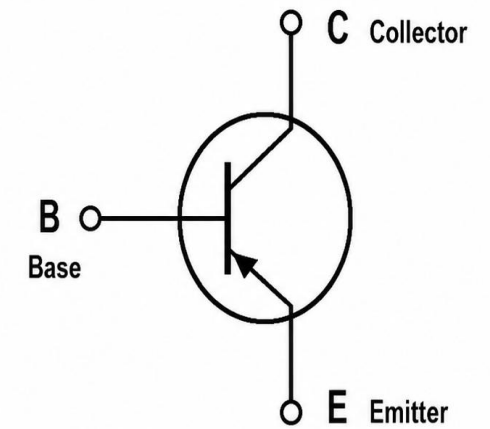
- A BJT is made of three semiconductor layers arranged as NPN or PNP
- It has three terminals: Emitter (E), Base (B), and Collector (C)
- The emitter is heavily doped to supply charge carriers
- The base is very thin and lightly doped
- The collector is moderately doped and physically larger to dissipate heat
- In an NPN transistor, the main charge carriers are electrons; in a PNP transistor, they are holes.

NPN BJT



N-P-N

PNP BJT

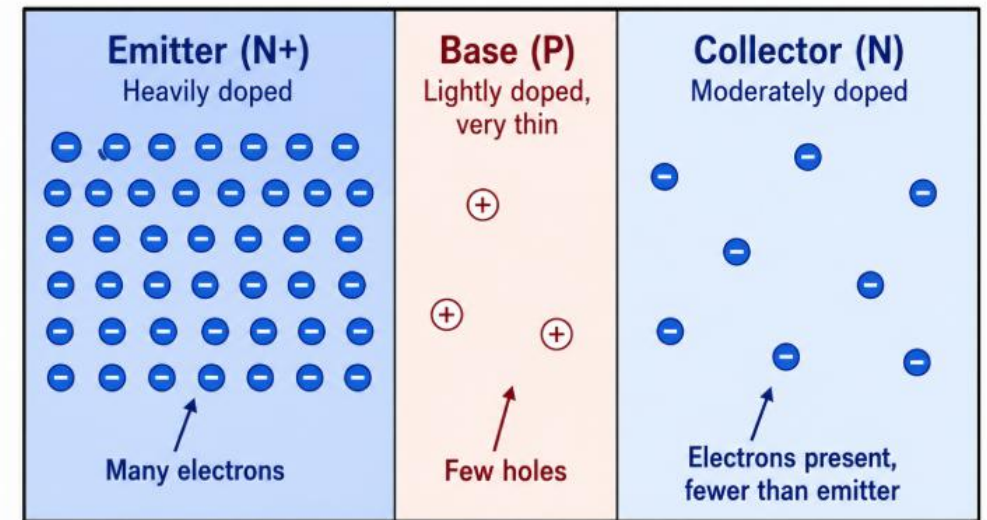


P-N-P

Working of BJT :

J1	J2	Region of operation	
F.B	R.B	Active	Amplifier
F.B	F.B	Saturation	ON state
R.B	R.B	Cut off	OFF state
R.B	F.B	Inverted	Rarely used

NPN BJT Structure

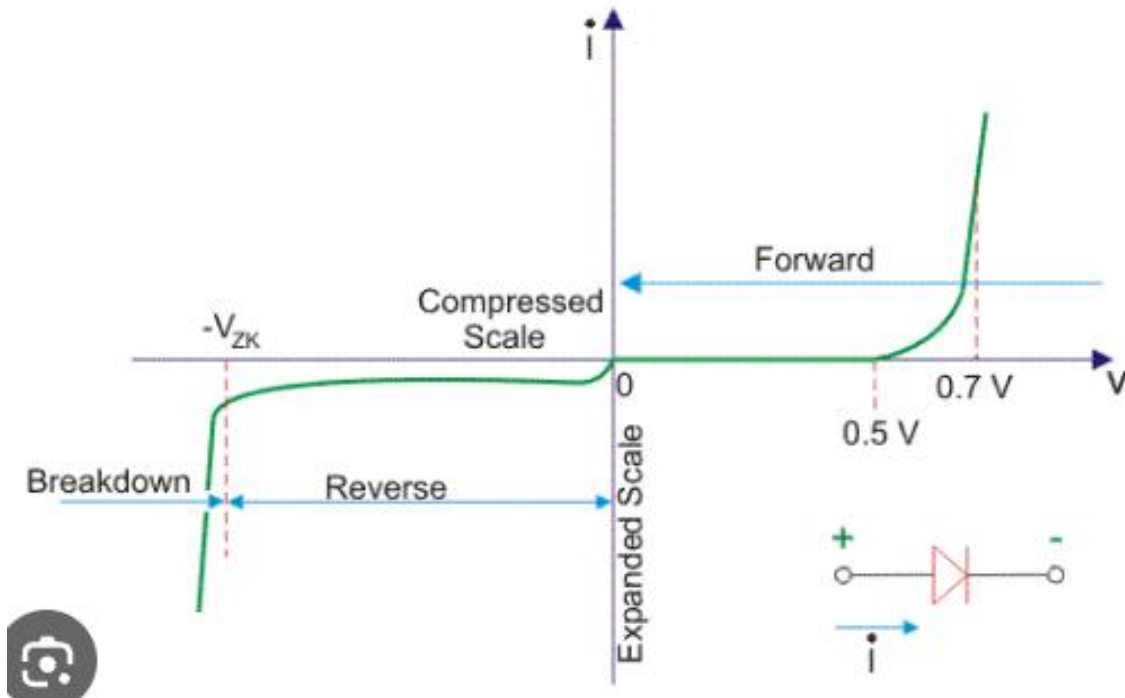


Emitter: highest doping
and most electrons

Base: light P doping
and few holes

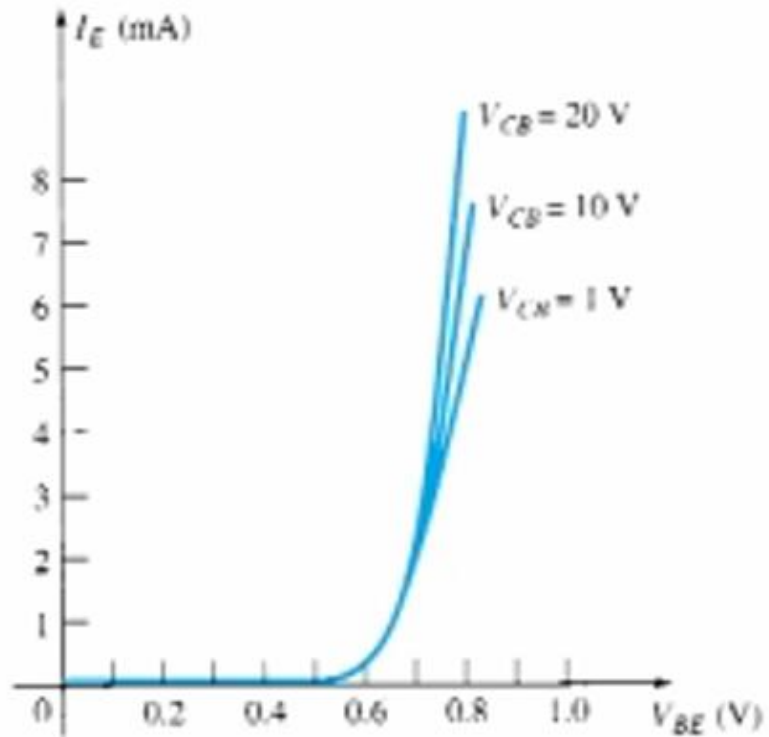
Collector: lower electron
concentration than emitter

Working of BJT :

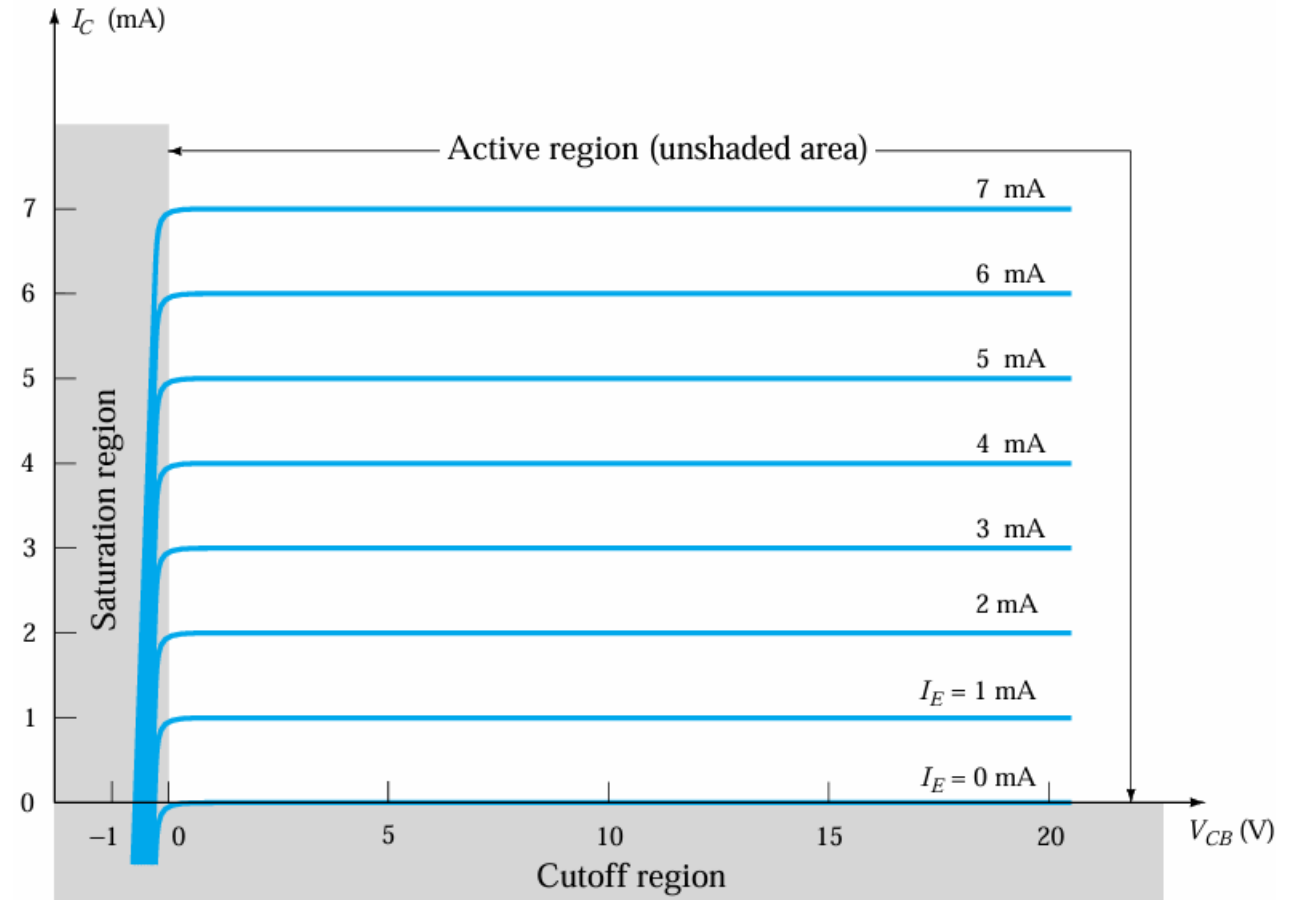


I-V curve of Diode

Working of BJT :



Input characteristic



Output characteristic

Transistor Amplifying Action :

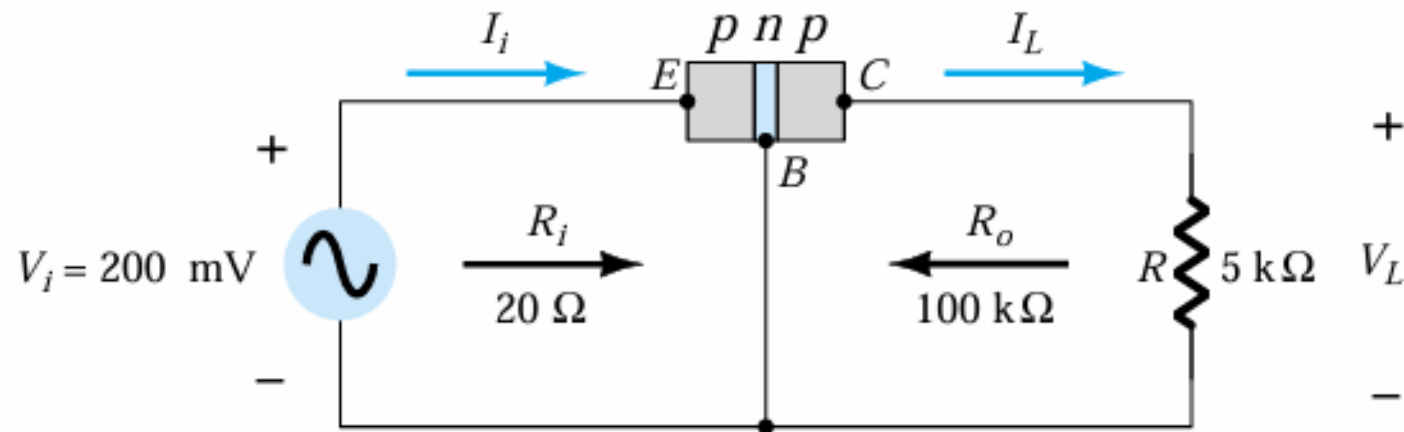
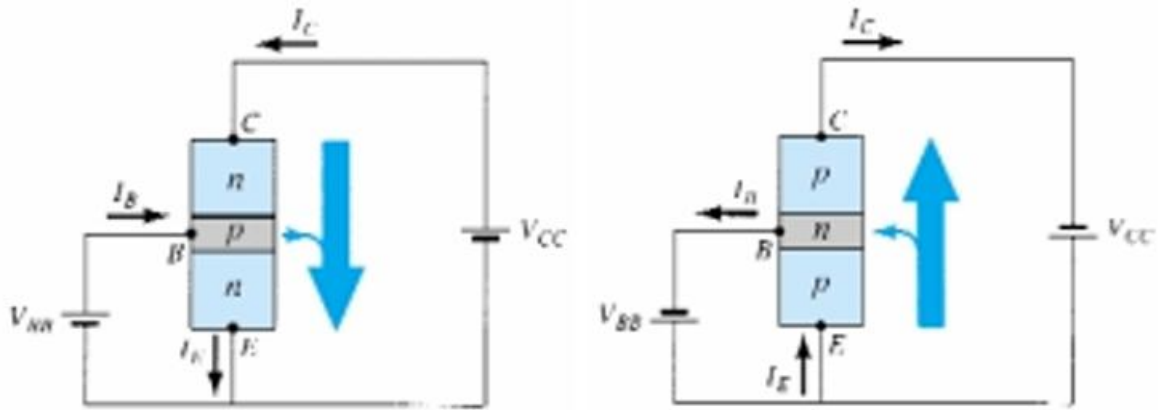
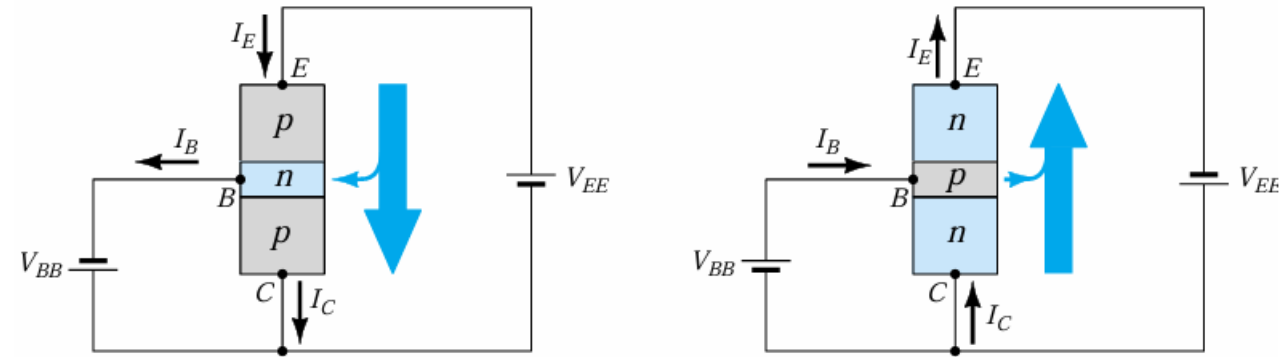


Figure 3.12 Basic voltage amplification action of the common-base configuration.

Other Configurations :



Common Emitter Configuration



Common Collector Configuration

THANK YOU ALL FOR
YOUR ATTENTION.